Dr Ali Moltajaei Farid

Senior Researcher — Swarm Robotics — Autonomous Systems — AI Control — Reinforcement Learning
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Professional Summary

Dedicated academic leader and senior robotics researcher with over 10 years of experience spanning international research institutions and high-tech industries. Proven expertise in Swarm Robotics, Multi-Agent Systems, and Autonomous UAV Control, with a track record of securing international funding (e.g., Mitacs, Petrobras) and publishing in high-impact journals. As a Lead Instructor and Supervisor, I have successfully mentored numerous MSc and PhD candidates, bridging the gap between theoretical reinforcement learning and real-world industrial deployments in precision agriculture and infrastructure. My research vision focuses on developing scalable, adaptive intelligence for heterogeneous robotic swarms to solve complex global challenges.

Education

PhD in Computer Science, Monash University, Australia, 2020
Thesis: *"An Investigation on Patrolling Swarms of UAVs"*
Key Achievements: Developed novel UAV patrolling strategies; explored battery displacement systems; hands-on work with Ardupilot, Pixhawk, ODROID XU3, Kinect, and E-Puck robots.

MSc in Electronics Engineering, University of Sistan & Baluchestan, Iran, 2013
Thesis: *"Designing Unmanned Aerial Vehicle Based on Neuro-Fuzzy Systems"* – Applied ANFIS to improve control in dynamic systems.

BSc in Control Engineering, Qazvin Azad University, Iran, 2010
Thesis: *"Designing Smart House Control System"* with sensor-based automation and anti-theft features.

Research Interests

- Reinforcement learning & multi-objective optimization
- Swarm robotics & multi-agent systems
- Intelligent control systems
- Robot path planning
- Neuro-fuzzy controllers
- Smart environments

Research & Professional Experience

Postdoctoral Researcher, Federal University of Paraiba (UFPB), Brazil, April 2026–Now
I am working on multi-UAV payload lifting supported by Petrobras.

Independent Researcher & Academic Supervisor, Oct 2024–March 2026
Conduct independent research on multi-robot systems and reinforcement learning. Supervise MSc and PhD students on reinforcement learning applications. Collaborated with leading companies on R&D of their products

Postdoctoral Researcher, K. N. Toosi University of Technology, Iran, 2023–Oct 2024
Designed multi-robotic precision farming systems using novel reinforcement learning methods to improve agricultural spraying.

Postdoctoral Researcher, University of Regina, Canada, 2021–2023
Led a collaborative project with Precision.ai on UAV swarms for weed control. Developed multi-UAV exploration and spraying algorithms.

Research & Development Engineer, Pand Industries, Iran, 2020–2021
Modernized existing industrial weighing devices and contributed to the development of next-generation smart models. Responsibilities included electronic circuit design and analysis, and microcontroller programming (ARM, PIC).

Research & Development Engineer, Maad Zist Fanavar Beynolmelal, Iran, 2019– 2021
Contributed to a dynamic R&D team focused on the development of advanced medical devices. Responsibilities included electronic circuit design and analysis, PCB layout design, algorithm development, and embedded programming (ARM, CPLD, PIC). Additionally, led efforts to document and comply with relevant medical standards, including risk management and safety protocols.

1 Research Supervision & Academic Mentorship

- **Co-Supervisor, Ph.D. Candidate: Fatemeh Khajemohammadi**, K. N. Toosi University of Technology
  - Project: Adaptive Controller Development using Reinforcement Learning.
  - Outcome: Directly guiding research methodology, resulting in high-impact manuscripts.
- **Co-Supervisor, Ph.D. Candidate: Abdulah Madadkar**, K. N. Toosi University of Technology
  - Project: Development of Novel Reinforcement Learning-based Exploration Strategies.
- **Research Mentor, MSc Student: Ali Reza Erfanian**, K. N. Toosi University of Technology
  - Project: Intelligent Guidance and Control of Quadcopters in Urban Environments based on Reinforcement Learning.
- **International Research Mentor, MITACS Globalink: Atharva Mete**, IIT Bombay & University of Regina
  - Project: Multiple Robot Motion Planning in a Dynamic Environment.
  - Outcome: Facilitated a successful international internship leading to a published conference paper at ICUAS 2023.
- **Academic Mentor, MSc Student: Md Samsul Arefin**, Gunma University, Japan
  - Project: Swarm UAVs for Search and Rescue Missions; related manuscript under review.

Publications

Journal Articles (Newest First)
1. Farid, A. M., Roshanian, J., Mouhoub, M. (2026). On-policy Actor-Critic Reinforcement Learning for Multiple Unmanned Aerial Vehicle Exploration. *Expert Systems with Applications*, 131496.
2. Khajemohammadi, F., Roshanian, J., Farid, A. M. (2025). A novel approach to tune fuzzy–PID membership functions based on reinforcement learning. *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering*
3. Khajemohammadi, F., Roshanian, J., Farid, A.M. (2025). Deep Reinforcement Learning-Based Optimization of an Innovative Fuzzy Inference System Structure for Quadrotor UAV Control. *Journal of Aerospace Science and Technology*, Published Sept 23, 2025.
4. Farid, A.M., Roshanian, J., Mouhoub, M. (2025). Multiple Aerial/Ground Vehicles Coordinated Spraying Using Reinforcement Learning. *Engineering Applications of Artificial Intelligence*, 151, 110686.
5. Farid, A.M. (2023). Effective UAV Patrolling for Swarm of Intruders with Heterogeneous Behavior. *Robotica*, 41(6), 1673–1688.
6. Farid, A.M., Egerton, S., Kamal, M.A.S. (2021). Search Strategies and Specifications in a Swarm versus Swarm Context. *Robotica*, 39(11), 1909–1925.
7. Farid, A.M., Barakati, S.M. (2013). UAV Controller Based on Adaptive Neuro-Fuzzy Inference System and PID. *International Journal of Robotics and Automation*, 2(2), 73–82.

Conference Papers (Newest First)
1. Farid, A.M., Roshanian, J., Mouhoub, M. (2026). On the Impact of LSTM Capacity on Recurrent Reinforcement Learning for Multi-UAV Exploration, *International Conference on Simulation and Modeling Methodologies, Technologies and Applications (SIMULTECH)*, Portugal.
2. Farid, A.M., Mouhoub, M. (2025). Online Collaborative UAV Path Planning for Mapping and Spraying Missions. *International Conference on Robotics, Computer Vision and Intelligent Systems*, Portugal.
3. Farid, A.M., Roshanian, J., Todorov, M., Serbezov, V., Reinforcement learning application in multi-UAV mapping and spraying(2025). *BulTrans 2024*, Bulgaria.
4. Farid, A.M., Mouhoub, M., Arkles, T., Hutch, G. (2024). Multi-UAV Weed Spraying. *International Conference on Robotics, Computer Vision and Intelligent Systems*, Italy.
5. Mete, A., Mouhoub, M., Farid, A.M. (2023). Coordinated Multi-Robot Exploration using Reinforcement Learning. *International Conference on Unmanned Aircraft Systems*, Poland.
6. Farid, A.M., Mouhoub, M. (2023). Multi-Objective Unmanned Aerial Mapping. *International Conference on Unmanned Aircraft Systems*, Poland.
7. Farid, A.M., Mouhoub, M. (2023). Nanobots in Medical Applications, a Future Horizon. *IEEE ICMAME*, UAE.
8. Farid, A.M., Mouhoub, M. (2022). Evolutionary Mapping with Multiple Unmanned Aerial Vehicles. *IEEE SMC*, Prague, Czech Republic.
9. Farid, A.M., Mouhoub, M., Sharifi, J. (2020). A Comparison on Intelligent Controllers based on Genetic Algorithms for Reducing Energy and Water Waste. *IEEE CEC*, Glasgow, UK.
10. Farid, A.M., Mouhoub, M., Sharifi, J., Barakati, S.M., Egerton, S. (2019). Multiple Objective Optimizers for Saving Water and Energy in Smart House. *IEEE SMC*, Bari, Italy.
11. Farid, A.M., Egerton, S., Barca, J.C., Kamal, M.A.S. (2018). Adaptive multi-objective search in a swarm vs swarm context. *IEEE SMC*, Miyazaki, Japan.
12. Farid, A.M., Egerton, S., Barca, J.C., Kamal, M.A.S. (2018). Search and Tracking in 3D Space Using a Species Based Particle Swarm Optimizer. *IEEE I2CACIS*, Shah Alam, Malaysia.
13. Ghansemi, N., Barakati, S.M., Farid, A.M. (2015). ANFIS Controller Based on RBF Iden-

tification for Piezoelectric Actuator in a Positioning System. *Iranian Joint Congress on Fuzzy and Intelligent Systems*, Zahedan, Iran.
14. Farid, A.M., Barakati, S.M. (2014). Implementation of DC Motor Neuro-Fuzzy Controller Using PSO Identification. *Iranian Conference on Electrical Engineering*, Tehran, Iran.
15. Farid, A.M., Barakati, S.M., Seifipour, N., Tayebi, N. (2013). Online ANFIS Controller Based on RBF Identification and PSO. *Asian Control Conference*, Istanbul, Turkey.

2 Selected Grants & Academic Awards

- **Petrobras Research Grant** (Postdoctoral Project) Apr 2026 – Present
  - Project: Multi-UAV Payload Lifting and Cooperative Control.
  - Awarded through the Federal University of Paraíba (UFPB), Brazil[cite: 26].
- **Iran's National Elites Foundation (INEF) Postdoctoral Funding** 2023 – 2025
  - Awarded for the project "UAV Swarm Spraying for Precision Agriculture"[cite: 80, 116].
- **Mitacs Accelerate Postdoctoral Fellowship**, Canada 2021 – 2023
  - Collaborative industrial-academic funding for UAV weed control projects with Precision.ai[cite: 79, 115].
- **Monash University Fully Funded PhD Scholarship**, Australia 2015 – 2020
  - Highly competitive full tuition and stipend award for doctoral research[cite: 77].
- **AMSI Internship Research Grant**, Australia 2015
  - Awarded by the Australian Mathematical Sciences Institute for specialized research integration[cite: 78].

Grants & Awards

- Monash University Fully Funded PhD Scholarship, Australia (2015–2020)
- Australian Mathematical Sciences Institute (AMSI) Internship Research Grant, Australia (2015)
- Canadian Mitacs Accelerate Postdoctoral Fellowship, Canada (2021–2023)
- Iran's National Elites Foundation Postdoctoral Funding, Iran (2023–2025)

Teaching Experience

K. N. Toosi University of Technology (KNTU)
- Main Instructor, *Artificial Intelligence Fundamentals*, Winter 2024 (In-person, 35 students)
- Main Instructor, *Algorithms and Programming Fundamentals*, Fall 2023 (In-person, 37 students)
- Main Instructor, *Algorithms and Programming Fundamentals*, Winter 2023 (In-person, 20 students)

University of Regina
- Main Instructor, *Advanced Algorithms and Data Structures (CS340)*, Summer 2022 (In-person, 44 students)
- Main Instructor, *Advanced Algorithms and Data Structures (CS340)*, Winter 2022 (Online, 43 students)

Monash University
- Teaching Assistant, *Theory of Computation (FIT2014)*, Winter 2018 (50 students)
- Teaching Assistant, *Object-Oriented Design and Implementation (FIT2099)*, Fall 2018 (40 students)

Supervision & Mentoring

- Mentored MSc student **Nima Ghasemi**, Mechatronics Engineering, project: ANFIS Controller Based on RBF Identification for a piezoelectric actuator in a Positioning System, resulting in a conference publication (2013–2014).

- Mentored BSc Globalink student **Atharva Mete**, Indian Institute of Technology Bombay and University of Regina, through the MITACS Globalink Research Internship. Project: Multiple Robot Motion Planning in a Dynamic Environment, resulting in a conference publication (May–July 2022).

- Co-supervised MSc student **Ali Reza Erfanian**, Aerospace Department, K. N. Toosi University of Technology, project: Intelligent Guidance and Control of a Quadcopter in an Urban Environment Based on Reinforcement Learning (from Jan 2024 to Nov 2025).

- Mentoring MSc student **Md Samsul Arefin**, Gunma University, Japan. Swarm UAVs for a search and rescue mission. Related manuscript currently under review.

- Co-supervising PhD student **Fatemeh Khajemohammadi**, Aerospace Department, K. N. Toosi University of Technology, project: Adaptive Controller Using Reinforcement Learning (from Jan 2024 to Jan 2026). Related manuscript currently under review.

- Co-supervising PhD student **Abdulah Madadkar**, Aerospace Department, K. N. Toosi University of Technology, project: A Novel Reinforcement Learning Based Exploration (from Nov 2025).

Academic Service

Thesis Committees

- Saman Yazdannik – MSc Defense, Cooperative Multi-Copter Control, Sept 2024

Peer Review Service

- IEEE Robotics and Automation Letters

- IEEE Transaction on Mobile Computing
- IEEE Access
- Robotica
- Springer J. Supercomputing
- Scientific Reports
- Drones

Grant Proposals

- **Canada:** MITACS UAV Weed Control (with Prof. Mouhoub)
- **Iran:** INEF UAV Swarm Spraying Project (with Prof. Roshanian)

Technical Skills

- Programming: Python, C++, Assembly, Matlab, ROS
- Electronics: AVR, ARM, PIC, STM32F4, Raspberry Pi, PCB Design (Altium, PROTEL)
- AI/ML: Reinforcement Learning, PyTorch
- Tools: Labview, Webots, Linux, LaTeX

Languages

- English – Full professional proficiency
- Persian, Turkish – Native/bilingual proficiency
- German, Arabic – Limited working proficiency

References

Prof. Malek Mouhoub – University of Regina – malek.mouhoub@uregina.ca Tel: +1-403-3804756
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